

Business Entity Resolution

Amazon ML Challenge 2026 · team briefing

What we are building, what the data says, how we intend to win,
and who does what across the three days.

Window	25 Sep 2026 00:00 IST -> 27 Sep 2026 23:59 IST
Uploads	5 per day, 15 total, then the submit button disables
Ranking	Private leaderboard decides the final result
Metric	F0.5 per Source-1 entity, macro-averaged over all entities
Team	P1 Abhinav (lead) · P2 blocking · P3 features · P4 decision and docs
Contract	AGENTS.md is binding. This document explains it; it does not replace it.

Every number in this document was measured from the dataset files on 25 Sep 2026. Nothing here is estimated or remembered.
Where a figure is not yet measured, it says so.

1. The problem

Three independent sources describe the same set of real businesses. They share no identifiers. Names and addresses are noisy: abbreviations, legal-suffix differences, transliterations, typos, missing components, landmark-based directions. **Source 1 is already deduplicated** and acts as the reference.

For **every** Source 1 record in the test set, we output which Source 2 and Source 3 records describe the same business. The answer may be zero, one, or many.

The deliverable is one tab-separated file, **1,732,544 rows plus a header**, one row per test Source 1 entity, with an empty field where the answer is "nothing matches this".

This is a record-linkage problem, and its defining difficulty is arithmetic: 1.7 million by 10 million is 17 trillion possible pairs. We cannot score them. So the pipeline must cheaply narrow the field, then judge what survives precisely. Everything else follows from that constraint.

Noise we must absorb

Field	Variation we have been told to expect
Name	Corp / Corporation, Pvt / Private, Ltd / Limited; DBA and trade names; & versus "and"; word-order transposition; typos
Address	Rd / Road, St / Street; transliteration variants; missing PIN or state; landmark references ("Near SBI ATM"); municipal numbering; reordering
Country	An open set of labels. Train has US and India. Test adds France. Never hard-code the set.

2. The data

Counted directly from the TSVs, not sampled or estimated.

	Train	Test
Source 1 entities	2,206,822	1,732,544
Source 2 records	5,034,616	4,887,273
Source 3 records	5,285,603	5,082,316
Countries (S1)	US 1,323,633 India 883,188	US 663,106 India 809,986 France 259,452 (15.0%)
Total size on disk	approx. 1.3 GB	approx. 1.2 GB

The five findings that shape our strategy

- **Singletons are 5.58% of entities** (123,247 of 2,206,822). Predicting an empty list for everything scores approximately 0.056. This matters enormously and is covered in the callout below.
- **Mean 3.67 true matches** per non-singleton entity; 7,638,365 true pairs in training. Most entities have several matches, and finding them is the work.
- **No Source 2 or 3 record belongs to two Source 1 entities.** Exactly zero across all 7.6 million pairs. One-to-one is a hard constraint we get for free, and it is a pure precision gain, which this metric rewards double.
- **France is 15.0% of the test set** and appears nowhere in training. Real, bounded, and the reason nothing in the pipeline may branch on a country name.

- **Test Source 2 and 3 records carry country labels, France included**, so blocking can group by country on every split without a fallback that would compare against all 10 million records.

3. The metric, and what it actually implies

F0.5 is computed **per Source 1 entity**, then averaged over **all** entities including singletons. An entity with one candidate counts exactly as much as an entity with forty. Pair-level accuracy, AUC and logloss are proxies only; none of them is the objective.

$$F0.5 = (1.25 \times P \times R) / (0.25 \times P + R)$$

Worked consequences, for an entity with 4 true matches

What we predict	Precision	Recall	F0.5
3 correct, 0 wrong	1.00	0.75	0.938
4 correct, 1 wrong	0.80	1.00	0.833
Nothing at all	-	0.00	0.000

Read those three rows together. Dropping an uncertain candidate beats adding a wrong one - that is the precision weighting. But refusing to answer at all is catastrophic, and that is the part a precision-first reflex gets wrong.

The correction that changes our strategy

Our own AGENTS.md originally said singletons were "a large share of the total score". They are 5.58%. Predicting empty everywhere scores 0.056, not something near 0.5. **94.4% of the available score requires actually finding matches.** The right posture is high precision that still commits on most entities - not maximum caution. A team that tunes toward silence caps itself near 0.056. AGENTS.md section 2 has been corrected; work from the corrected version.

4. How the pipeline works

Five stages, all in src/. Each one narrows or sharpens what the next one sees.

Stage	File	What it does	Why it matters
1. Normalise	normalize.py	Lowercase, strip accents so Societe matches Société, expand abbreviations, split the name into full and legal-suffix-free "core" forms.	Country-agnostic by construction. This is what makes an unseen France survivable.
2. Block	blocking.py	Three TF-IDF views - char n-grams on the core name, char n-grams on name plus address, word n-grams on the address - take the nearest few per entity, union them. Grouped by country.	Sets the ceiling on everything. A true match dropped here can never be recovered downstream.
3. Pair features	pair_features.py	28 features per surviving pair: fuzzy ratios, Jaro-Winkler, TF-IDF cosines, postal and number agreement, plus rank and competition context.	The context features let the model reason about alternatives, not just one pair in isolation.
4. Match	run_pipeline.py	LightGBM, 5 folds grouped by Source 1 entity, predicts P(this pair is a match).	MIT licensed. Fast enough to retrain repeatedly, which matters more than raw ceiling over three days.

Stage	File	What it does	Why it matters
5. Decide	decide.py	Threshold the probabilities, then enforce one-to-one so each Source 2 or 3 record is claimed by at most one entity.	Where the score is actually won. The only stage that directly sets precision, and currently the least invested in.

5. How we intend to win

Priority order, highest leverage first

#	Lever	Why it ranks here
1	Make blocking run at full scale	It is the ceiling on every other number, and today the implementation cannot finish. Until this lands we have no submission at all, so its expected value dominates everything else on the board.
2	The decision layer: threshold plus a real one-to-one assignment	Precision is doubly weighted and this is the only stage that sets it directly. The one-to-one constraint is exactly true in training, and we currently exploit it only greedily. Cheap to improve, measurable in one run.
3	Feature quality	Real but incremental. Worth doing once stages 1 and 2 are solid, and only with cross-fitted evidence behind each change.
4	France robustness	15% of the score. The pipeline is already country-agnostic, so this is insurance against a regression rather than a source of upside.

The disciplines that protect the score

- **Report only cross-fitted numbers.** `tune()` picks its threshold on the same rows it scores, so it flatters itself. `cross_fitted_score()` is the honest one, and it is the only number that belongs in STATUS.md.
- **Accept a change only if** it beats seed noise by more than 2x **and** improves at least 4 of 5 folds. Anything smaller is logged as inconclusive and dropped. Most ideas will not clear this bar, and that is the point.
- **Score every entity, including the ones blocking found nothing for.** An entity with no candidates is a real prediction of "empty" and still scores. Filtering them out inflates the number and fools us.
- **Trust CV over the public leaderboard.** The public board scores a subset; the private board decides the result. When they disagree, suspect a submission bug first, then a metric mismatch, then leakage, then shift. Noise is the last explanation, never the first.
- **Check LOCO on anything that touches features.** A change that lifts CV but drops leave-one-country-out will hurt us on 15% of the test set.

How we spend 15 uploads

Uploads answer questions cross-validation cannot. They do not settle threshold nudges, hyperparameter tweaks or curiosity - a public-board difference smaller than its own noise is not evidence of anything.

Worth an upload	Never worth an upload
The format check (the empty submission) The first working end-to-end pipeline A structurally different blocking or feature family An ensemble The two reserved final picks	A threshold nudge A hyperparameter tweak A blend-weight change "Let's just see what happens" Anything we could have answered with CV

6. Team workflow

Who	Owns	Current task
P1 Abhinav Team lead	metric.py, cv_folds.py, run_pipeline.py, STATUS.md, all uploads, final zip	Issue #5. Runs the pipeline, owns every upload, generates and commits the locked folds file.
P2 Data analyst	normalize.py, blocking.py	Issue #1 - critical path. Replace the dense top-k with a sparse one so blocking can finish.
P3 ML engineer	pair_features.py, model params	Issue #2. Vectorise feature construction and remove the pair-length reindex.
P4 Decision and docs	decide.py, approach_document.md, STATUS.md header	Issues #3 and #4. Decision layer, and start the methodology document on day 1.

Rules of the road

- Branch per person. Small commits. Merge to main fast - **self-merge plus a chat ping beats waiting for review**. With 68 hours left, a pull request sitting unreviewed overnight is the expensive failure, not an unreviewed merge.
- Before any push: `metric.py` passes, and your change ran end to end at least once.
- **Never run** `git add .` - name the paths. A blanket add has already pulled an entire virtual environment into a commit once.
- Always use `.venv\Scripts\python`, never bare `python`. Build the environment with Python 3.12; 3.14 cannot install our pinned versions.
- After every experiment: one row in STATUS.md section 5, one line in section 8. Section 8's headings match the organisers' template, so the write-up assembles itself instead of being written from scratch at midnight on day 3.
- Every upload gets a git tag and a ledger row **before** it goes up. Version history is required for shortlisting.
- Develop against `--sample 2000` for a three-minute feedback loop. Its scores are not CV and must never be logged as results.

The rule that ends our run if broken

No external data lookup of any kind. No geocoding APIs, no business registries, no commercial entity-resolution services, no scraped or downloaded reference data. Hand-written abbreviation dictionaries are fine - they rewrite tokens already in the record. A list of real place names would not be: that is a gazetteer, which is external data. Code packages are reviewed by humans, and the penalty is disqualification for the whole team regardless of who wrote the line.

7. The three days

Each day has one job. A day that achieves its job and nothing else is a good day; a day that produces five interesting experiments and no working pipeline is not.

Day 1 (25 Sep) - make it run

Job: turn code that has never executed into a pipeline that finishes. Nothing else on this list matters until blocking and features scale.

- P2 and P3 work issues #1 and #2 in parallel. These are the critical path.
- P1 uploads D1-1, the empty submission. It proves the upload path end to end and its public score reveals the singleton share of the public subset - information we cannot obtain any other way.
- P1 sends the organisers' query form the one question that matters: does the private leaderboard score our last submission, our best, or one we select? Neither official document states it, and it decides the entire day 3 endgame.
- P4 starts the methodology document and the decision-layer work against sampled output.
- **Day 1 done when:** a full-scale run has been started, and D1-1 is on the board.

Day 2 (26 Sep) - make it good

Job: produce the first honest CV number, then improve it deliberately.

- First full-scale run completes. P1 records blocking recall, cross-fitted CV and LOCO in STATUS.md, and commits work/folds.csv so every teammate shares the same folds from that point on.
- First real model submission. Compare its public score against CV and reconcile any gap before trusting either.
- P4 takes the decision layer seriously: proper one-to-one assignment, threshold behaviour by candidate count. This is the highest-value modelling work available.
- Feature and blocking iterations, each one accepted or rejected on the 2x-noise and 4-of-5-folds rule.
- **Day 2 done when:** we have a cross-fitted CV number we believe, and at least one real model submission scored on the public board.

Day 3 (27 Sep) - make it safe

Job: convert whatever we have into the best defensible submission, and stop in time. Day 3 is where good scores get lost to rushing.

- Morning: last structural experiment. Anything not working by midday is abandoned, not rescued.
- Afternoon: freeze the pipeline. Final full run. Validate with --check-ids.
- **By 22:00 IST:** both reserved uploads in - the best-CV version, and one hedge that differs on our biggest open risk, France robustness. Do not leave these to the last hour; the portal is busiest then.
- **By 23:00 IST:** final zip ready - both TSVs, a runnable code folder with README and pinned requirements, and the filled methodology document.
- **Day 3 done when:** the zip would survive a reviewer opening it cold, without us in the room to explain anything.

Inside any day

Pull main. Take everyone's work. Do one thing. Run it. Log the number - whether it helped or not, because an inconclusive result recorded is worth more than a promising one forgotten. Merge. Ping the chat. Repeat.

Post in chat when you start something, when you finish, and immediately when you are blocked. That last one is what actually saves hours. Nobody is watching GitHub notifications at three in the morning.